

Stochastic Gradient Descent

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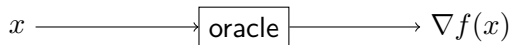




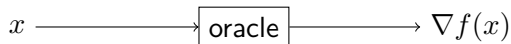
1. Motivation for SGD
2. Stochastic Gradient Descent
3. Examples



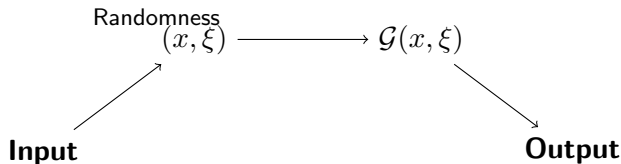
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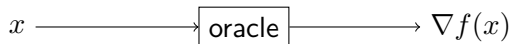
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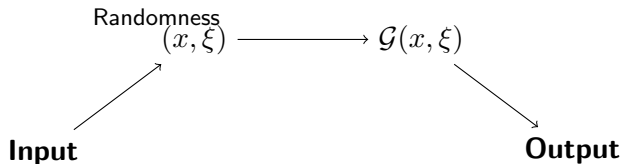
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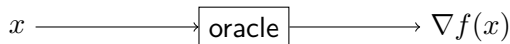
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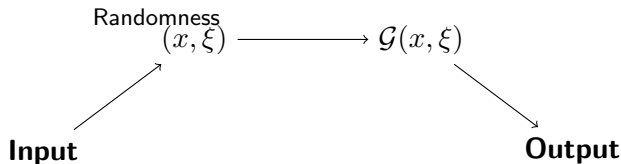
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In large scale machine learning, using **SGD** is often a better choice due to computational efficiency.



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$$f(x) = \frac{1}{n} \sum_{i=1}^n \log \left(1 + e^{b_i a_i^\top x} \right) \quad [\text{Binary classification}]$$



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With this, define:

$$\mathcal{G}(x; \xi) = \nabla f_{\xi}(x) \quad (\text{Random variable because of } \xi)$$



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- **Quality:** Questionable
- **Quality of $\mathcal{G}(x, \xi)$ to be controlled.**(variance control)





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$$\mathbb{E}_{\xi} \left[\|G(x; \xi)\|^2 \right] \leq B^2$$



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Back to Finite Sum Problem





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- **Unbiasedness:** $\mathbb{E}_\xi[W(\xi)] = 0$
- **Boundedness assumption:** Assume f is L -Lipschitz, then describe the boundedness assumption (**Exercise!**)

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where $e_\xi^\top = [0 \ \dots \ 1 \ \dots \ 0]$ is the standard basis vector.



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- **Boundedness:** **Exercise!** If f is L -Lipschitz, then

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Remark: This is useful, since we expect $\|x - x^*\|$ to be small, implying $\mathbb{E}_{\xi} [\|G(x; \xi)\|]$ is small.

Thank You!